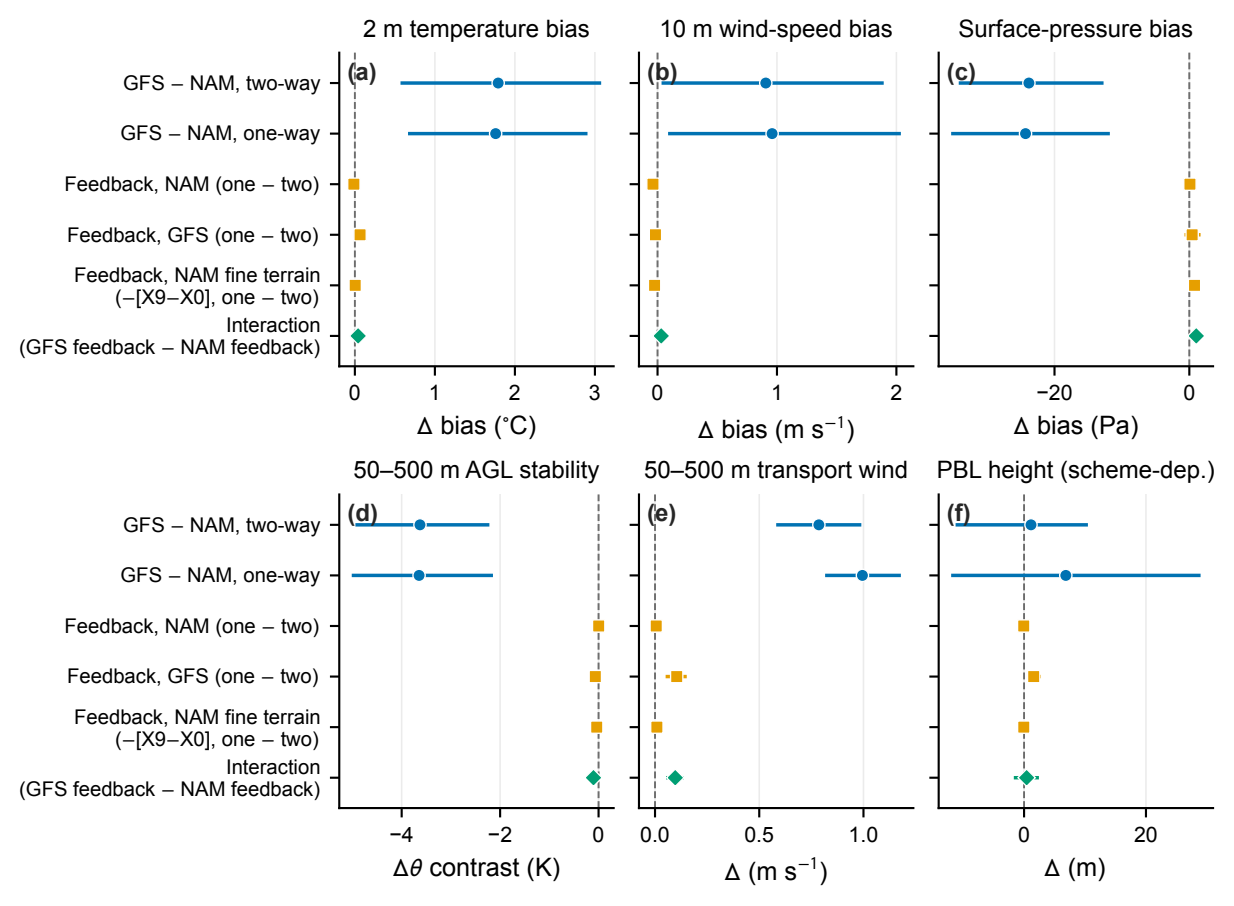

2 m temperature bias
10 m wind-speed bias
Surface-pressure bias
(a)
(b)
(c)
GFS – NAM, two-way
GFS – NAM, one-way
Feedback, NAM (one – two)
Feedback, GFS (one – two)
Feedback, NAM fine terrain
(–[X9–X0], one – two)
Interaction
(GFS feedback – NAM feedback)
0
1
2
3
0
1
2
−20
0
Δ bias (°C)
Δ bias (m s−1)
Δ bias (Pa)
50–500 m AGL stability
50–500 m transport wind
PBL height (scheme-dep.)
(d)
(e)
(f)
GFS – NAM, two-way
GFS – NAM, one-way
Feedback, NAM (one – two)
Feedback, GFS (one – two)
Feedback, NAM fine terrain
(–[X9–X0], one – two)
Interaction
(GFS feedback – NAM feedback)
−4
−2
0
0.0
0.5
1.0
0
20
Δθ contrast (K)
Δ (m s−1)
Δ (m)